

Notes: Jovanovic (Econometrica, 1982)

Selection and the Evolution of Industry

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1 Big picture

- **Question.** Why do small firms grow faster, display more volatile growth, and fail more often than large firms? The paper builds a theory in which these facts arise from selection and learning rather than from a proportional-growth law.
- **Core idea.** Firms enter without knowing their own efficiency. They learn from their realized costs. Favorable signals lead firms to expand and survive; unfavorable signals lead them to shrink and exit.
- **Economic mechanism.** Each firm is a draw from a distribution of true costs. Low-cost firms gradually discover that they are efficient, produce more, and remain in the industry. High-cost firms receive bad cost signals, reduce output, and eventually exit.
- **Market structure.** Firms are infinitesimal and face a deterministic equilibrium price path. There is idiosyncratic uncertainty at the firm level but no aggregate uncertainty, so aggregate output and prices are deterministic in equilibrium.
- **Entry and exit.** Entry requires a sunk cost. Once a firm enters, it chooses output while learning about its true cost. Exit occurs when the value of continuing falls below the value of using the firm's fixed factor elsewhere.
- **Main theoretical result.** A perfect-foresight equilibrium exists, is unique, and coincides with the unique maximizer of discounted consumer plus producer surplus.
- **Main economic predictions.** The model predicts that young and small firms have higher and more variable growth rates, lower survival probabilities, and stronger selection effects. Mature surviving firms become larger and more stable.
- **Economic takeaway.** Firm-size dynamics need not come from fixed adjustment costs or exogenous firm-specific growth rates. They can arise endogenously from Bayesian learning about heterogeneous productivity and the exit of inefficient firms.

2 Incremental contribution of the paper

- **From proportional growth to selection.** Earlier theories often rationalized Gibrat-style proportional growth, where firms grow at rates unrelated to size. This paper shows how systematic departures from proportional growth can arise from noisy selection.
- **A learning model of firm dynamics.** The paper embeds Bayesian learning into firm entry, output, and exit decisions. Firm size is not just an initial condition; it is an outcome of accumulated information about efficiency.
- **A theory of firm exit.** Adjustment-cost models can generate smooth growth but usually do not generate endogenous failure. Jovanovic makes exit central: inefficient firms decline and leave the industry.
- **A bridge between micro evidence and industry evolution.** The model connects plant- or firm-level patterns - small firms grow faster and fail more - to industry-level outcomes such as concentration, average profitability, and entry.

- **A selection interpretation of profitability patterns.** The paper explains why profits and concentration may rise together as an industry matures: low-profit firms exit, leaving a more efficient survivor population.
- **Equilibrium contribution.** The paper does not only solve a single firm's stopping problem. It embeds that problem in a perfect-foresight industry equilibrium with entry and endogenous prices.
- **Welfare and uniqueness contribution.** The proof strategy shows that equilibrium is the unique maximum of a discounted surplus problem. This gives the model a strong competitive-equilibrium interpretation despite the complicated state space.
- **Foundation for later firm-dynamics research.** The paper provides a canonical mechanism - noisy learning plus selection - that later models of firm entry, survival, and growth build on.

3 Evidence, primitives, and state variables

3.1 Target facts

- The paper is motivated by evidence that small firms grow faster than large firms and are more likely to fail.
- Small-firm growth is also more variable. This matters because high average growth among surviving small firms may partly reflect selection out of failing firms.
- The paper also speaks to broader industry facts: concentration and rates of return may be positively related, persistence in returns may be higher for large firms, and concentrated industries may exhibit more dispersion in returns.
- These facts conflict with a simple proportional-growth benchmark in which all firms grow at the same rate and failure is absent.

Interpretation: The model is not designed to fit one narrow statistic. It is designed to explain a cluster of facts about growth, survival, concentration, and profitability using one mechanism: learning about efficiency.

3.2 Cost technology and unknown type

- Output is homogeneous. A firm producing q has production cost

$$c(q)x_t,$$

where $c(0) = 0$, $c'(0) = 0$, $c'(q) > 0$, $c''(q) > 0$, and marginal cost eventually becomes arbitrarily large.

- The cost shifter x_t is random. It depends on a latent firm type θ and an idiosyncratic signal:

$$\eta_t = \theta + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2) \text{ iid},$$

with $x_t = \xi(\eta_t)$ for an increasing transformation $\xi(\cdot)$.

- A higher θ means higher expected costs and therefore lower efficiency.

- Potential entrants share a common prior over θ . A firm does not know whether it is truly efficient when it enters.

Interpretation: The key heterogeneity is permanent but initially hidden. A firm is not born knowing whether it is good or bad; it discovers this through noisy operating experience.

3.3 Learning and sufficient statistics

- At age n , the firm's belief about θ is summarized by the number of observations n and the sample mean of cost signals $\bar{\eta}_n$.
- Equivalently, the firm's state can be summarized by (x_t^*, n) , where

$$x_t^* = E[x_t \mid \text{information before } t]$$

is expected current cost conditional on the firm's information.

- The sequence x_t^* is a martingale: before the next signal arrives, the firm's best forecast of future expected cost is its current expected cost.
- As the firm ages, posterior uncertainty falls. Learning is most important for young firms.

Interpretation: Firm age matters because it measures information precision. A young firm with the same current expected cost as an old firm still has more uncertainty and therefore more volatile future growth.

3.4 Outside value and entry cost

- The firm has an outside value $W > 0$, interpreted as the value of the fixed factor - for example managerial ability or a location - if used outside the industry.
- The outside value is assumed not to rise one-for-one when the firm learns that it is efficient in the current industry.
- Entry requires a sunk cost k .
- The value of entry at date t is $V(x_0, 0, t; p) - k$, where x_0 is the prior expected cost and p is the anticipated price sequence.

Interpretation: The outside value gives exit a real opportunity cost interpretation. A firm exits not because output hits zero mechanically, but because continuing in this industry is no longer better than redeploying its fixed factor elsewhere.

4 Firm problem and individual dynamics

4.1 Output choice

Given expected cost x_t^* and price p_t , the firm chooses output before observing the current cost shock. Let

$$q_t = q\left(\frac{p_t}{x_t^*}\right).$$

The paper shows that output is decreasing in expected cost:

$$\frac{\partial q}{\partial x_t^*} = -\frac{c'}{x_t^* c''} < 0.$$

- Good news lowers expected cost x_t^* .
- Lower expected cost raises output.
- Bad news raises expected cost and reduces output.

Interpretation: Firm size is the empirical shadow of beliefs. A firm that grows is not simply lucky in output demand; it has learned that it is likely to have low costs.

4.2 One-period profit

Expected one-period profit, after optimizing over output, is

$$\pi(p_t, x) = p_t q(p_t/x) - c(q(p_t/x)) x.$$

- Profits are higher when expected cost x is lower.
- Realized high profits are informative because they imply that realized cost was low relative to expected cost.
- This creates a link between current profitability and future growth: unexpectedly high profits lead the firm to revise expected cost downward, which raises future output.

Interpretation: The model gives a learning-based reason why high current profits predict subsequent growth. Profits are not only internal finance; they are information.

4.3 Continuation value and exit rule

For a bounded price sequence $p = \{p_t\}_{t=0}^\infty$, the value of staying in the industry for one period and then behaving optimally is

$$V(x, n, t; p) = \pi(p_t, x) + \beta \int \max\{W, V(z, n+1, t+1; p)\} P(dz \mid x, n).$$

The exit cutoff $\gamma(n, t; p)$ solves

$$V(\gamma(n, t; p), n, t; p) = W.$$

- Since V is decreasing in x , a high expected cost makes exit more attractive.
- If $x_t^* > \gamma(n, t; p)$, the firm exits.
- The corresponding output threshold is $q[p_t/\gamma(n, t; p)]$: firms whose output path falls below the boundary leave.

Interpretation: Exit is a stopping rule based on beliefs. Inefficient firms do not necessarily exit immediately; they wait until accumulated evidence makes the continuation value too low.

4.4 Selection and survivor bias

- Firms that survive tend to be those whose signals lowered expected costs.
- Failing firms are small at the time of failure, because high expected costs imply low output.
- If datasets omit failed firms, measured small-firm growth is biased upward because the small firms that would have grown slowly or declined have exited.
- The paper emphasizes that even after accounting for this bias, small firms should still have higher and more variable growth because learning is stronger for young firms.

Interpretation: The model separates two forces: true selection, which changes the composition of survivors, and statistical selection bias, which appears when failed firms disappear from samples.

5 Industry equilibrium

5.1 Industry supply

Let y_τ be the measure of firms entering at date τ . For firms of vintage τ , let $\phi(t, \tau; p)$ denote expected output at date t conditional on optimal production and exit behavior under the price sequence p . Aggregate industry output is

$$Q_t(p, y) = \sum_{\tau=0}^t y_\tau \phi(t, \tau; p).$$

- Each individual firm is infinitesimal, but there is a continuum of firms.
- Idiosyncratic firm-level uncertainty averages out in the aggregate.
- Aggregate supply is deterministic for a given price path and entry sequence.

Interpretation: This is how individual learning and exit become an industry supply curve. The output of each vintage depends on the distribution of surviving expected costs.

5.2 Perfect-foresight equilibrium

Let $D(Q_t, t)$ be the inverse demand function. A perfect-foresight equilibrium consists of output rules, exit rules, a price sequence p , and an entry sequence y such that

$$p_t = D(Q_t(p, y), t),$$

and

$$\begin{aligned} V(x_0, 0, t; p) - k &= W & \text{if } y_t > 0, \\ V(x_0, 0, t; p) - k &\leq W & \text{if } y_t = 0. \end{aligned}$$

- The price sequence is self-fulfilling: firms optimize taking it as given, and their aggregate behavior generates exactly that price sequence.
- Entry has a zero-profit condition when entry occurs.
- If entry value is below the outside value, no firms enter.

Interpretation: Entry is the margin that pins down industry evolution. If current and future prices make entry too attractive, more firms enter until the entry value is driven back down.

5.3 Planner problem and social surplus

The paper proves equilibrium existence and uniqueness by showing that equilibrium solves a surplus-maximization problem. Define

$$S(Q) = \sum_{t=0}^{\infty} \beta^t \int_0^{Q_t} D(z, t) dz - K(Q),$$

where $K(Q)$ is the minimized present-value cost of producing the aggregate output sequence Q .

- The planner faces the same information structure as firms.
- The planner chooses entry, exit, and output assignments to produce an aggregate output path at least cost.
- The key proof difficulty is that the state of the industry is a distribution over firm expected costs and ages.

Interpretation: The welfare proof is not just a technical appendix. It shows that decentralized entry, exit, and learning do not create inefficient timing of entry or exit in this competitive setting.

6 Main theoretical results

6.1 Theorem 1: the firm value function

- The Bellman equation for $V(x, n, t; p)$ has a unique bounded continuous solution.
- The value function is strictly decreasing in expected cost x .
- The monotonicity of V makes the exit cutoff well defined.

Economic message: Better expected efficiency increases the value of staying. Therefore the model produces a clean sorting rule: firms with sufficiently bad posterior beliefs exit.

6.2 Theorem 2: existence, uniqueness, and optimality

- There is exactly one equilibrium price sequence and one entry sequence satisfying the equilibrium conditions.
- The equilibrium allocation is the unique maximum of discounted consumer plus producer surplus.
- The equilibrium price can be interpreted as the social marginal cost associated with the planner's output sequence.

Economic message: Competitive selection is efficient in this model. Firms do not enter too early, exit too late, or produce the wrong quantities, because there are no externalities and prices fully coordinate decentralized decisions.

6.3 Theorem 3: constant prices and continuing entry

Theorem 3 assumes that demand is nondecreasing over time and that $q(p/x)$ is strictly concave in x . Under these conditions:

- the equilibrium price sequence is constant;
- entry occurs every period;
- the value of entry is exactly driven to the entry condition in every period.

Economic message: When surviving firms' output tends to fall as uncertainty is resolved and exit occurs, growing demand is met by new entry rather than by a rising price path. Industry growth therefore contains both selection among incumbents and continuing inflow of new firms.

6.4 Theorem 4: homogeneous demand shifts

- The minimized cost functional $K(Q)$ is homogeneous of degree one.
- An equiproportional rightward shift in demand raises equilibrium output in the same proportion at all dates.
- Equilibrium prices are unchanged.

Economic message: Pure scale changes in demand affect how many firms enter and how much the industry produces, but not the price path. This reflects the constant factor prices and the continuum of potential entrants.

7 Empirical implications and interpretation

7.1 Firm growth and Gibrat's law

- The proportional-growth law says growth rates should be independent of size.
- Jovanovic's model predicts that young and small firms should have higher growth rates because their posterior beliefs are less precise and favorable news can sharply reduce expected costs.
- It also predicts that their growth rates should be more variable.
- Among mature survivors, beliefs converge and growth becomes less variable; under special cost functions, mature firms can approximately satisfy proportional growth.

Interpretation: The paper reconciles two empirical patterns: small firms violate Gibrat's law, while samples of large mature firms may look closer to proportional growth.

7.2 Survival and exit

- The firms most likely to exit are those whose posterior expected costs rise.
- Exit is concentrated among smaller firms because high expected cost maps into low output.
- At the time of failure, the failing firm is smaller than the surviving firms in its cohort.

Interpretation: Failure is not an exogenous shock. It is the market's way of removing firms whose histories reveal that they are inefficient producers.

7.3 Concentration and profitability

- Firms begin with similar beliefs, but posterior expected costs diverge over time.
- This divergence creates dispersion in firm size and raises measures of concentration such as the Gini coefficient.
- As inefficient firms exit, average profits among survivors rise, especially if prices do not fall.
- High concentration can therefore be associated with high profits for large firms, but not necessarily for small marginal firms.

Interpretation: Concentration is not only a market-power story. In this model, concentration partly reflects successful selection: high-rent firms are those revealed to be efficient.

7.4 Current profits and future growth

- A high realized profit for a given firm size indicates that realized cost was unusually low.
- This causes a downward revision in expected cost.
- Lower expected cost implies higher future output.
- Thus unusually high current profits predict unusually high future growth.

Interpretation: The model gives an informational channel from profits to growth. This is distinct from a financing channel in which high profits matter only because they relax borrowing constraints.

7.5 Entry versus incumbent expansion

- If $q(p/x)$ is concave in expected cost, the output of a surviving vintage tends to decline when prices do not rise.
- Under growing demand and constant prices, the additional demand must be supplied by new entrants.
- The industry can therefore grow even while many incumbent firms exit and individual cohorts shrink.

Interpretation: Aggregate industry growth can coexist with heavy firm turnover. The model separates the dynamics of a cohort from the dynamics of the whole industry.

8 Identification and empirical design

8.1 What the paper identifies theoretically

- The paper identifies a mechanism that can jointly generate size-dependent growth, growth volatility, and exit.
- It shows that these patterns do not require adjustment costs, decreasing returns at the firm level beyond convex production costs, or exogenous firm-specific growth-rate heterogeneity.

- The central identifying logic is comparative statics: more uncertain and smaller firms should have stronger learning-driven growth and failure patterns.

Interpretation: This is a theory paper, not a direct empirical estimation. Its contribution is to discipline a set of stylized facts with a coherent equilibrium model.

8.2 What the model abstracts from

- There is no aggregate uncertainty: demand is deterministic and known.
- Firms are infinitesimal, so strategic interaction and market power are absent.
- The product is homogeneous.
- Capital adjustment frictions are not the central mechanism.
- The outside value W is common and does not move one-for-one with information about industry-specific efficiency.

Interpretation: These assumptions make the selection channel transparent. They also limit the model's ability to address strategic entry deterrence, aggregate recessions, financing constraints, or multiproduct firms.

8.3 How to read the predictions empirically

- A clean empirical test should track entrants over time, including failures, because omitting exits overstates small-firm growth.
- The model predicts stronger growth variance and failure among young firms, not just small firms mechanically.
- The model also predicts that profitability contains information about future growth after conditioning on size.
- Industry-level concentration should be interpreted partly as the outcome of selection, not only as evidence of anti-competitive behavior.

Interpretation: The paper tells empirical researchers to pay attention to survival bias, age, and the information content of profits when studying firm growth.

9 Figures and tables

9.1 Figure 1: Firm output, survival, and failure

Read:

- Figure 1 is the paper's main visual summary. It shows output over time for firms entering at date τ .
- The upward solid path is a survivor. Its output rises as the firm receives favorable cost signals and remains in the continuation region $V > W$.
- The dashed path is a failing firm. Its output declines as the firm receives unfavorable signals and eventually crosses into the exit region.

- The lower curve $q[p_t/\gamma(t-\tau, t; p)]$ is the output threshold implied by the exit cutoff. Below this boundary, continuation value is no greater than outside value.
- The figure makes the key selection mechanism transparent: efficient firms grow and survive; inefficient firms decline and fail.

Economic message: The figure turns the Bellman equation into an economic story. Growth is the observable consequence of good information, and exit is the observable consequence of bad information. The paper has one formal figure and no empirical tables, so this single figure is displayed at width 0.6 as requested.

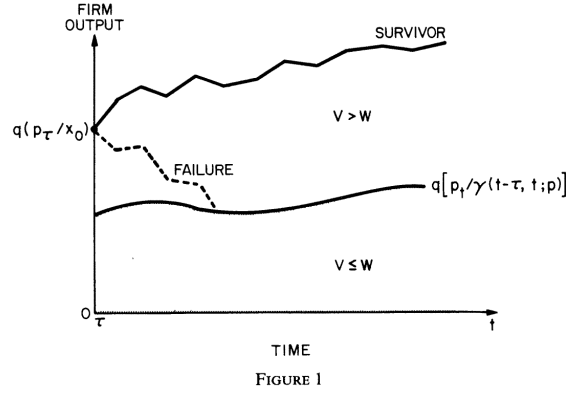


Figure 1: Output paths, survival, failure, and the endogenous exit boundary.

10 Short summary

- The paper develops a model of industry evolution driven by selection and learning.
- Firms differ in true efficiency but do not know their own type at entry.
- Operating experience generates noisy signals about efficiency.
- Firms with favorable signals update expected costs downward, expand output, and survive.
- Firms with unfavorable signals update expected costs upward, shrink, and exit once continuation value falls below the outside value.
- The model explains why small firms grow faster, have more volatile growth, and fail more often.
- It also explains why survivor profitability and concentration may rise as an industry matures.
- The equilibrium is unique and efficient in the competitive setting.
- The main conceptual contribution is to make selection, rather than adjustment costs alone, a core driver of firm-size dynamics.

11 Conclusion

11.1 Core conclusion

Jovanovic shows that industry evolution can be understood as a process of noisy selection. Firms enter uncertain about their own efficiency, learn from cost realizations, expand if the news is good, and leave if the news is bad. This simple learning mechanism generates rich firm dynamics.

11.2 Economic intuition

The model's intuition is powerful because it links beliefs directly to output. A firm's expected cost summarizes what it has learned. Lower expected cost means higher output and a higher continuation value. Higher expected cost means lower output and a higher probability of exit. Over time, the market selects toward firms whose histories reveal them to be efficient.

11.3 Incremental contribution revisited

The paper moves the literature beyond a proportional-growth benchmark. It explains why small firms grow faster and fail more without assuming that small firms have mechanically higher growth opportunities. They are simply less tested. Some turn out to be highly efficient and grow rapidly; others reveal themselves to be inefficient and exit.

11.4 Implications for empirical work

Empirical studies of firm growth should distinguish size from age, account for failure, and treat profits as signals about underlying productivity. A sample that includes only surviving firms can misstate the growth process because it omits precisely the firms whose information turned bad.

11.5 Limitations

The model abstracts from aggregate shocks, financial frictions, strategic competition, multiproduct firms, and heterogeneous demand conditions. It is therefore best read as a benchmark theory of competitive selection rather than a complete model of every industry cycle.

11.6 Takeaway

The key takeaway is that firm growth and survival are information problems. Markets learn which firms are efficient, and industry structure evolves because the efficient firms grow while the inefficient firms disappear.